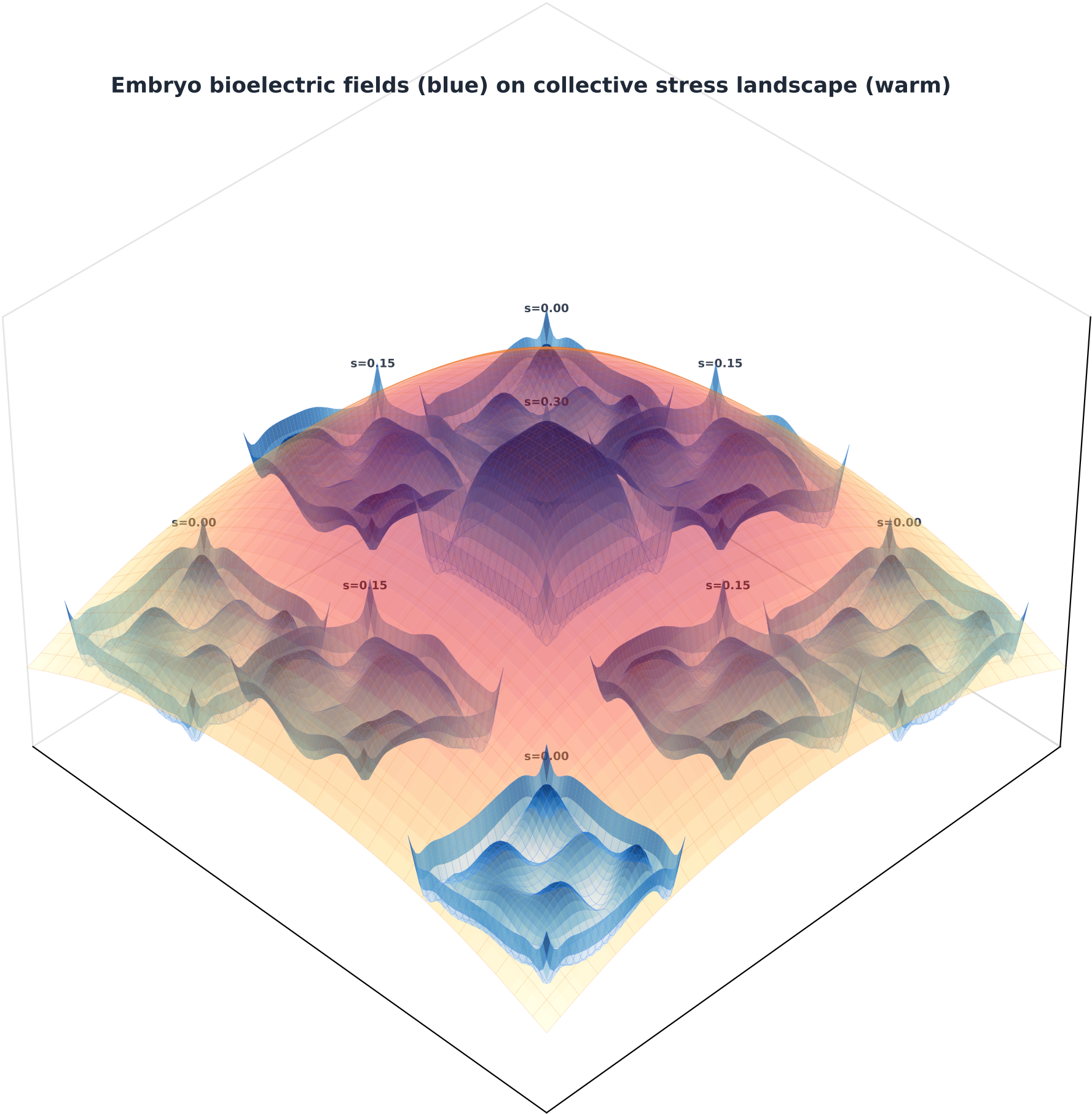


MULTI-EMBRYO STRESS RESCUE NETWORK

Embryo bioelectric fields (blue) on collective stress landscape (warm)



Embryo Stress Level

- Healthy ($s = 0.0$)
- Mildly stressed ($s = 0.15$)
- Strongly stressed ($s = 0.30$)

Stress Field Equation

$$\frac{dF}{dt} = D_F \nabla^2 F - \gamma_F F + \text{emission}$$

$\text{emission} = \text{mean}(S) \text{ per embryo}$

Rescue Mechanism

Stressed embryos emit eATP into the shared medium. Collective signal reshapes each receiver's ATP dynamics via saddle-node bifurcation.